

CS & IT ENGINEERING



Computer Network - 2

Transport & Application Layer

Lecture No. - 06



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Recap of Previous Lecture



Topic

DNS ✓





Topics to be Covered



Topic

HTTP

$\Rightarrow$ V. Imp.



ABOUT ME



Hello, I'm **Abhishek**

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Topic : Web Page



→ Web page consists of objects

[Object can be HTML file, JPEG image, Java Applet, Audio File etc]

→ Each objects can be stored on different web servers

→ Web page consists of base HTML file which includes several referenced objects

[Each addressable by a URL]

HTML File :-

- Hyper text Markup lang
- text File (.html)
- Browser $\Rightarrow$ Rendering Process





Topic : HTTP



- Hypertext Transfer Protocol (HTTP)
- 'Stateless' Application Layer Protocol
- Client - Server Model



Topic : HTTP



Client :

→ Browser that ^①requests, ^②receives, (using HTTP protocol)
and ^③display web objects

Server :

→ Web server sends (using HTTP protocol) objects in response to requests



Topic : HTTP uses TCP

Version 1 & 2



✓ → Server creates TCP socket on Port 80 (default port number)

→ Client initiates TCP connection to server ✓

→ Server accepts TCP connection from client ✓

DNS = 53
HTTP = 80

3-way handshake ←

TCP connection Establishment



Topic : HTTP uses TCP



→ HTTP messages (application layer protocol messages) exchanged
between browser (HTTP client) and web server (HTTP server)

→ TCP connection closed

→ TCP Data Transfer
[After connection
Establish]

* HTTP communication



Topic : HTTP Connections



→ Two types of HTTP connections :

1. Non-persistent HTTP : HTTP/1.0
2. Persistent HTTP : HTTP/1.1

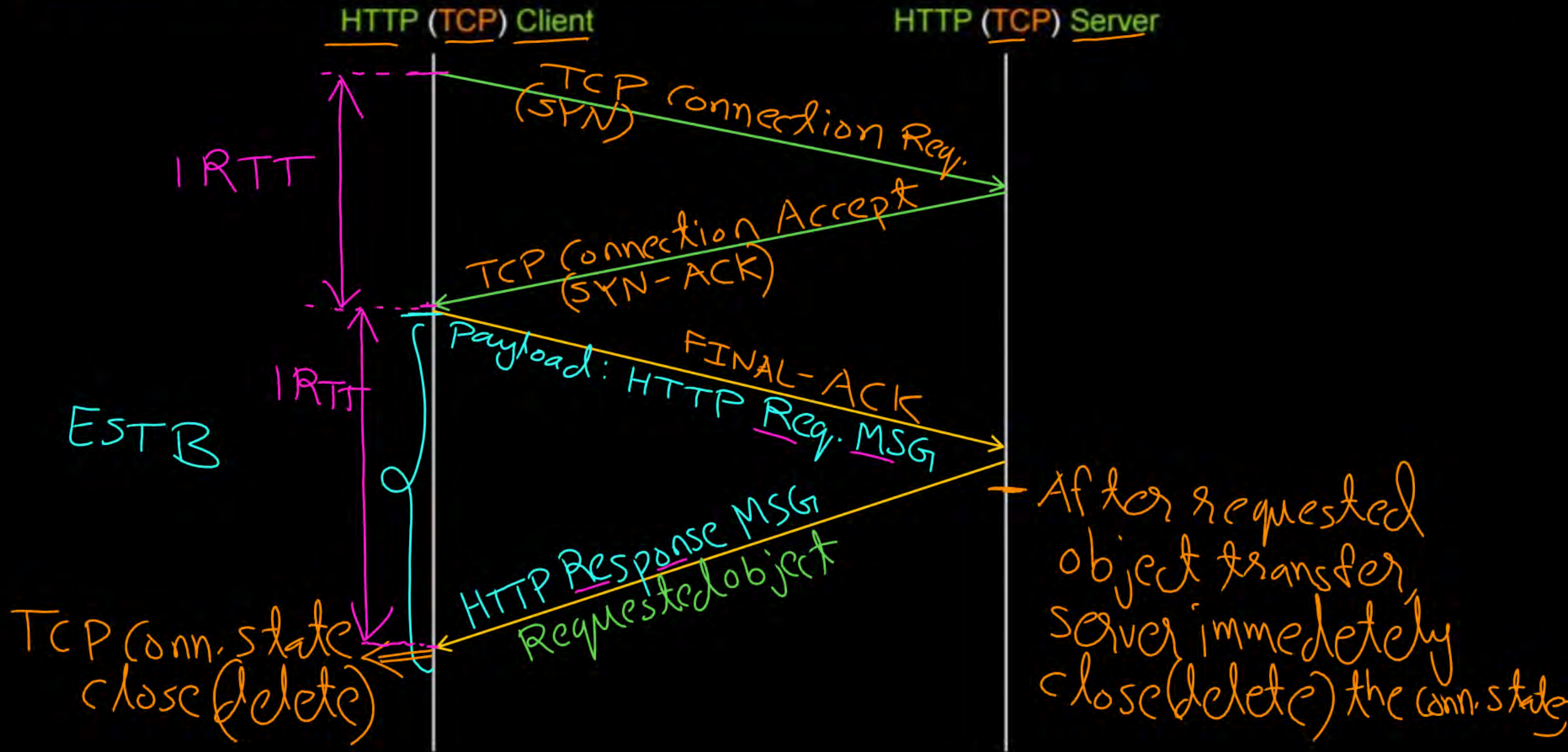


Topic : Non-persistent HTTP (HTTP/1.0)

- At most one object sent over a TCP connection
- Downloading multiple objects required multiple TCP connections



Topic : Non-persistent HTTP (HTTP/1.0)



Example 3 :-



#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP. The web page contains base HTML file and ten very small image objects. How many TCP connections required to load the web page completely?

→ Non-persistent HTTP

⇒ one-TCP connection : Base HTML File

Refer ⇒ 10 URL(s)

* one-TCP connection for each Refered object.
(separate)

Total = 11 TCP connections

Ans = 11

[NAT] [GATE-2020, 1-Mark]

IIT-D, H.W.



#Q. Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is _____.



Topic : Non-persistent HTTP (HTTP/1.0)

HTTP response time (per object) :

- One RTT to initiate TCP connection
- One RTT for HTTP request and first few bytes of HTTP response to return
- Object / File transmission time

$$\text{Response time} = (2 * \text{RTT}) + \text{File Transmission Time}$$

⇒ For very small size objects, object(file) transmission time can be negligible.

Example 4 :-



#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP (no any parallel TCP connections). The web page contains base HTML file and ten very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?

⇒ Non-persistent HTTP (No any parallel TCP conn.)

① Base HTML File = $1 \text{ RTT} + 1 \text{ RTT} = 2 \text{ RTT}$

→ Base HTML File transmission time is always negligible.

② For each referenced object = $1 \text{ RTT} + 1 \text{ RTT} = 2 \text{ RTT}$

→ As per question, object transmission time are negligible.

Total = 22 RTT

Ans = 22



Topic : Non-persistent HTTP (HTTP/1.0)

- Requires 2 RTT per object
- Operating System overhead for each TCP connection
- Browser often open multiple parallel TCP connection
to fetch referenced objects in parallel

Example 5 :-



#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP with five parallel TCP connections. The web page contains base HTML file and ten very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?

→ Non-persistent HTTP (with five parallel TCP conn.)

① 2 RTT : Base HTML File.

② 2 RTT : five image object using 5 parallel TCP conn.

③ 2 RTT : Next five - // - // - // - // - //

Total = 6 RTT

Ans = 6



Topic : Persistent HTTP (HTTP/1.1)

→ Server leaves connection open after sending response

→ Subsequent HTTP messages between same client/server sent over open connection (TCP)

→ Multiple objects can be sent over single TCP connection
[between Client and Server] [if all the objects are placed in same Server]

⇒ At the end, 4-way handshake needed to close the connection;



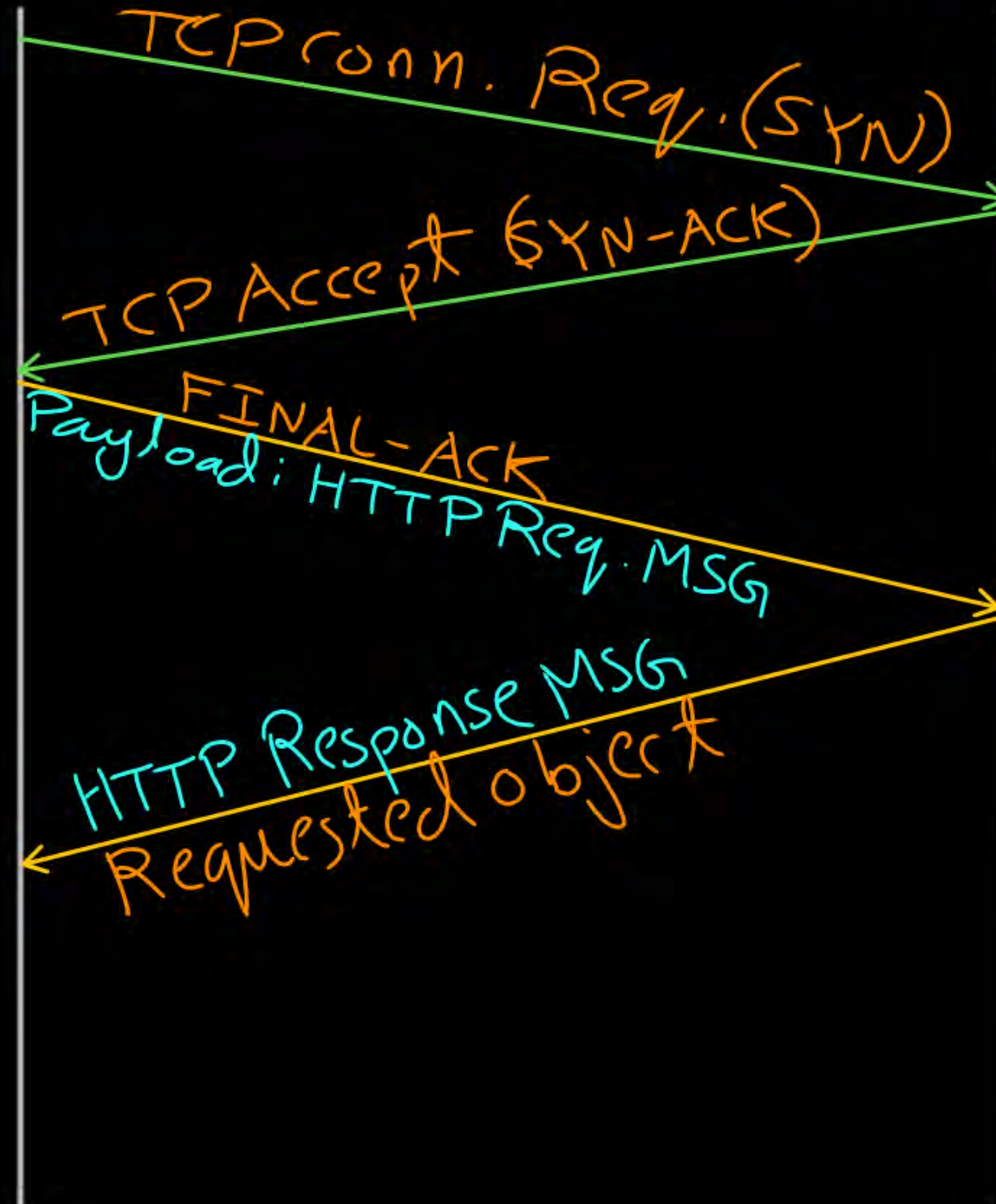
Topic : Persistent HTTP (HTTP/1.1)

→ onwards



HTTP (TCP) Client

HTTP (TCP) Server



Example 6 :-

#Q. Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP. The web page contains base HTML file and ten very small image objects. Assume base HTML file and all ten image objects are placed in same web server. How many TCP connections required to load the web page completely?

Ans = 1

Example 7 :-



#Q. Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP. The web page contains base HTML file and ten very small image objects. Assume base HTML file and all ten image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?

→ Persistent HTTP (without pipeline)

① one-RTT : TCP conn.

② one-RTT : Base HTML File.

③ one-RTT for each object.

$$\text{Total} = 12 \text{ RTT} = 1 \text{ RTT} + 1 \text{ RTT} + 10 \text{ RTT}$$

Ans = 12

[MCQ] [GATE-2014, Set-2, 2-Marks] IIT-KGP, H.W.



#Q. A graphical HTML browser resident at a network client machine Q accesses a static HTML webpage from a HTTP server S. The static HTML page has exactly one static embedded image which is also at S. Assuming no caching, which one of the following is correct about the HTML webpage loading (including the embedded image) ?

- A** Q needs to send at least 2 HTTP requests to S, each necessarily in a separate TCP connection to server S
- B** Q needs to send at least 2 HTTP requests to S, but a single TCP connection to server S is sufficient
- C** A single HTTP request from Q to S is sufficient, and a single TCP connection between Q and S is necessary for this
- D** A single HTTP request from Q to S is sufficient, and this is possible without any TCP connection between Q and S



Topic : Persistent HTTP (HTTP/1.1)

→ Client sends requests as soon as it encounters a referenced object (Non-pipeline)

→ One RTT for each referenced objects (instead of two)

→ After Reusing the open TCP connection



Topic : Persistent HTTP (HTTP/1.1)

- Client can make requests in pipeline ✓
- Server can send referenced objects in pipeline ✓
- One RTT for all referenced objects (for small size objects)

Example 8 :-



#Q. Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP with pipelining. The web page contains base HTML file and ten very small image objects. Assume base HTML file and all ten image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?

→ Persistent HTTP (with pipeline):-

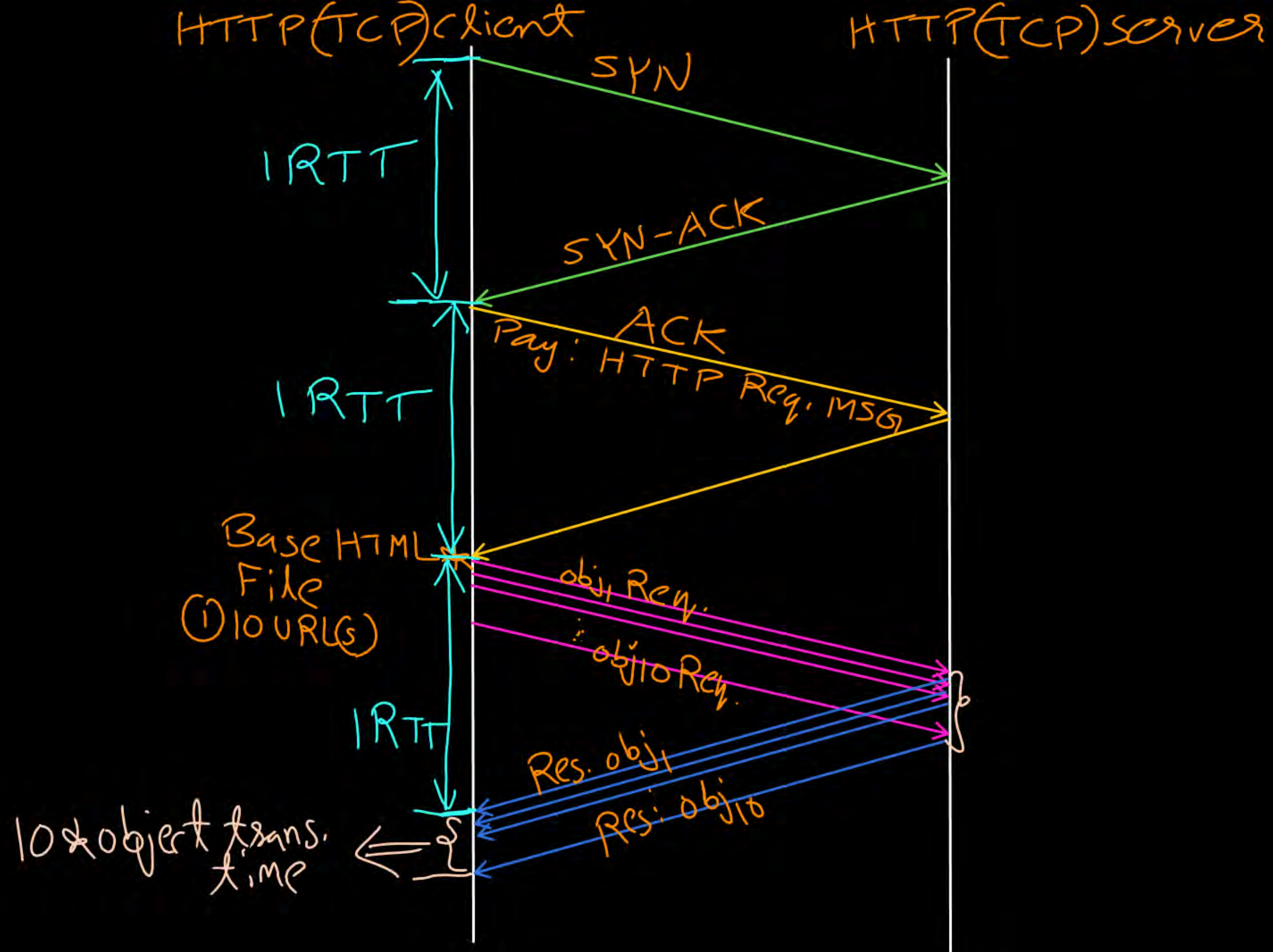
① one-RTT: TCP conn.

② one-RTT: Base HTML File

③ one-RTT: All 10 objects in pipeline

Total = 3 RTT

Ans = 3



[MSQ] [GATE-2023, 2-Marks] IIT-K / H.W.



#Q. Suppose in a web browser, you click on the `www.gate-2023.in` URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers.

Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting `www.gate-2023.in` is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server.

Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

- A** 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
- B** 5 RTTs, in case of persistent HTTP with pipelining.
- C** 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
- D** 6 RTTs, in case of persistent HTTP with pipelining.

[MCQ] [GATE-2016, Set-2, 1-Marks] ISC, H.W.



#Q. Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

- A** HTTP GET request, DNS query, TCP SYN
- B** DNS query, HTTP GET request, TCP SYN
- C** DNS query, TCP SYN, HTTP GET request
- D** TCP SYN, DNS query, HTTP GET request

[MCQ] [GATE-2024, Set-1, 1-Mark] ISC, H.W



#Q. A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

- (i) HTTP GET request for the index page
- (ii) DNS request to resolve the web server's name to its IP address
- (iii) HTTP GET request for an image object
- (iv) TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer?

- | | |
|---------------------------------|---------------------------------|
| A (iv), (ii), (iii), (i) | B (ii), (iv), (i), (iii) |
| C (iv), (ii), (i), (iii) | D (ii), (iv), (iii), (i) |



2 mins Summary



Topic

HTTP

⇒ Email
SMTP
IMAP/POP
⇒ FTP



THANK - YOU

